Quadruped - Autonomous Four-Legged Walking Robot

by Robotics Society

Abstract:

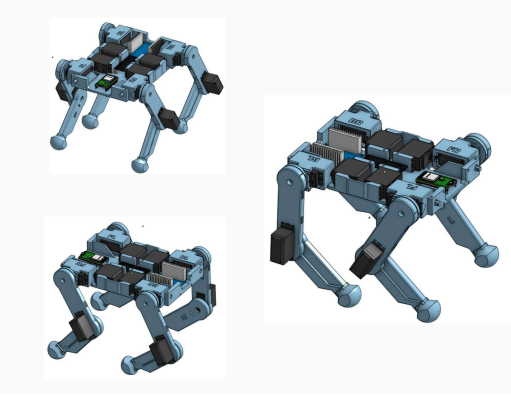
This project develops a four-legged walking robot capable of stable and coordinated locomotion. It uses inverse kinematics and gait planning to control individual leg movements and generate different walking patterns. The project focuses on developing a compact and modular platform for studying quadruped locomotion, robotics and control systems.

Approach:

The robot uses four independently controlled legs with three degrees of freedom per leg. Inverse kinematics converts the desired foot position into joint angles, while gait planning coordinates the movement of all four legs. Different swing and stance trajectories are generated to achieve smooth and stable motion.

Design:

The quadruped is designed as a lightweight four-legged robotic platform with three servo-driven joints per leg: hip, upper-leg and lower-leg joints. The mechanical structure is designed symmetrically to provide balanced weight distribution and a wide range of foot movement. The system uses an ESP32 as the main controller and a PCA9685 driver for controlling the servos.



Applications:

Robotics Research: Study of locomotion, gait generation and control.
Education: Practical learning of robotics, mechanics and programming.
Terrain Navigation: Platform for experimenting with legged movement on uneven surfaces.
Inspection: Potential use in environments unsuitable for conventional wheeled robots.

Future Aspects/Goals:

1. Active stabilization for improved balance.
2. Adaptive gait generation for different terrains and speeds.
3. Autonomous navigation using environmental sensors.
4. Advance locomotion including climbing and obstacle traversal.

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Pictures & Videos

